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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CYS102 – CYBER FORENSICS

(For CYS)

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Module**.

Module – I

	Marks	BT*	CO*	PO*
1. a) Describe the auditing and security architecture in Windows operating system with a suitable diagram.	7	2	1	1
b) Discuss the hives of windows registry in detail.	7	2	1	1
c) What are the different types of metadata of windows operating system? Differentiate between FAT and NTFS file system.	6	3	1	1
2. a) With a suitable diagram, discuss the fundamentals of end point forensics for Microsoft windows - Kernel and device driver architecture.	7	2	1	1
b) With a neat diagram explain how data is stored and retrieved from the hard disk.	7	2	1	1
c) Explain the concept of slack space in computer forensic with appropriate diagram.	6	2	1	1

Module – II

3. a) What is device driver? Illustrate kernel and device driver architecture of Linux operating system with a suitable diagram.	7	2	2	1
b) List and explain the several different pieces of Linux.	7	2	2	1
c) Explain the features of EXT2/3/4 file systems.	6	2	2	1
4. a) Discuss the key security and audit mechanism in Linux.	10	2	2	1
b) Illustrate the process involved in Reconstruction of file and directory structures using UFS.	10	2	2	1

Module – III

5. a) What is database forensic? Discuss Importance of Database Forensics in Cybersecurity.	10	2	3	1
b) Illustrate the steps involved in performing the forensics on database.	10	2	3	1
6. a) Why indexing is essential in Table Storage? Demonstrate the concept of primary and secondary indexing in Table storage.	10	3	3	1
b) Discuss the steps involved in recovering the database if it is deleted.	10	2	3	1

Module – IV

7. a) Illustrate the network components and discuss the role of each OSI layer in facilitating data flow in the network with appropriate diagrams.	10	2	4	1
b) What are the different Types of Wireless Networks in Forensics? How to collect key evidence in Wireless Forensics? Discuss in detail.	10	2	4	1
8. a) Explain the different types of web attack and also Illustrate how to conduct forensic investigation during these web attacks.	10	2	4	1



	b) What is Social Media forensics? Explain the various stages and tools applicable in each stage in the process of social media forensics	10	2	4	1
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Module – V

9.	a) Outline the features of the following Tools (i) FTK Imager (ii) Oxygen Forensic Detective (iii) Autopsy	10	2	5	1
	b) Discuss the different types of offenses and sections under ITA act 2000	10	2	5	1
10.	a) What is DOS attack? Explain the different types or levels of DOS attack and also list the tools used to launch the DOS attack.	10	2	5	1
	b) What are the survival (5P) Mantra for Netizens? Discuss any two case studies with respect to cybercrime.	10	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
